

Supplementary material: Optimal radiotherapy dose scheduling with variable fraction sizes and breaks via sequential mixed-integer convex programming, *Physics and Imaging in Radiation Oncology*

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May 12, 2026

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A Derivation of Problem 9

We are interested in formulating the dose scheduling problem as a finite-horizon optimal control problem with the dose fractions as the control variables (we directly manipulate these variables to effect changes in the biological system) and the cell counts and cell compartment configuration as the state variables (representing all potential states of the biological system that are relevant to our problem). Assuming a fixed schedule length, our goal is to minimize the tumor cell survival fraction (SF) at the end of treatment subject to an upper bound on the total normal tissue BED₃, as given by expressions 1 and 2. The SF and BED₃ are connected to the dose schedule through the cell counts, which evolve according to the dynamics described in Section B. In order to formulate our optimization problem, we must represent these cell dynamics as a set of constraints that define the mathematical relationship between the dose fractions and the cell counts in each compartment.

A.1 Pre-recompartmentalization

We begin by combining the cell dynamics given by equations S25—S30, which characterize how cells evolve prior to recompartmentalization:

$$\tilde{N}_t^{P,v} = N_t^{P,v} \exp \left(-\alpha_P d_t - \beta_P d_t^2 + f_{pro}^P \frac{\log 2}{T_C} \Delta t \right) \quad (S1)$$

$$\tilde{N}_t^{P,d} = (N_t^{P,d} + N_t^{P,v} (1 - \exp(-\alpha_P d_t - \beta_P d_t^2))) \exp \left((2k_m - 1) f_{pro}^P \frac{\log 2}{T_C} \Delta t \right) \quad (S2)$$

$$\tilde{N}_t^{I,v} = N_t^{I,v} \exp(-\alpha_I d_t - \beta_I d_t^2) \quad (S3)$$

$$\tilde{N}_t^{I,d} = N_t^{I,d} + N_t^{I,v} (1 - \exp(-\alpha_I d_t - \beta_I d_t^2)) \quad (S4)$$

$$\dot{N}_t^{H,v} = N_t^{H,v} \exp(-\alpha_H d_t - \beta_H d_t^2) \quad (S5)$$

$$\dot{N}_t^{H,d} = N_t^{H,d} + N_t^{H,v} (1 - \exp(-\alpha_H d_t - \beta_H d_t^2)) \quad (S6)$$

$$\tilde{N}_t^{H,v} + \tilde{N}_t^{H,d} = (\dot{N}_t^{H,v} + \dot{N}_t^{H,d}) \exp \left(-\frac{\log 2}{T_{1/2,loss}} \Delta t \right) \quad (S7)$$

Here $N_t = (N_t^{P,v}, N_t^{P,d}, N_t^{I,v}, N_t^{I,d}, N_t^{H,v}, N_t^{H,d})$ are the cell counts at the beginning of time step t , $\dot{N}_t = (\dot{N}_t^{H,v}, \dot{N}_t^{H,d})$ are the H compartment cell counts after radiation cell-kill, but before hypoxic cell loss or recompartmentalization in time step t , and $\tilde{N}_t = (\tilde{N}_t^{P,v}, \tilde{N}_t^{P,d}, \tilde{N}_t^{I,v}, \tilde{N}_t^{I,d}, \tilde{N}_t^{H,v}, \tilde{N}_t^{H,d})$ are the cell counts after radiation cell-kill, proliferation, hypoxic cell loss, and mitotic cell death, but before recompartmentalization in time step t , for $t = 1, \dots, T$.

For simplicity, define the constants

$$c_1 = \exp \left(f_{pro}^P \frac{\log 2}{T_C} \Delta t \right), \quad c_2 = \exp \left((2k_m - 1) f_{pro}^P \frac{\log 2}{T_C} \Delta t \right), \quad c_3 = \exp \left(-\frac{\log 2}{T_{1/2,loss}} \Delta t \right)$$

We take logs of both sides of equation S1 to get

$$\log \tilde{N}_t^{P,v} = \log \left(N_t^{P,v} c_1 \exp(-\alpha_P d_t - \beta_P d_t^2) \right) = \log N_t^{P,v} - \alpha_P d_t - \beta_P d_t^2 + \log c_1 \quad (S8)$$

By plugging equation S1 into S2, we obtain a linear representation

$$\tilde{N}_t^{P,d} = c_2 N_t^{P,d} + c_2 N_t^{P,v} - c_2 N_t^{P,v} \exp(-\alpha_P d_t - \beta_P d_t^2) = c_2 N_t^{P,d} + c_2 N_t^{P,v} - \frac{c_2}{c_1} \tilde{N}_t^{P,v} \quad (S9)$$

The I and H compartment dynamics can be reduced in a similar manner to

$$\log \tilde{N}_t^{I,v} = \log N_t^{I,v} - \alpha_I d_t - \beta_I d_t^2 \quad (\text{S10})$$

$$\tilde{N}_t^{I,d} = N_t^{I,d} + N_t^{I,v} - \tilde{N}_t^{I,v} \quad (\text{S11})$$

$$\log \dot{N}_t^{H,v} = \log N_t^{H,v} - \alpha_H d_t - \beta_H d_t^2 \quad (\text{S12})$$

$$\dot{N}_t^{H,d} = N_t^{H,d} + N_t^{H,v} - \dot{N}_t^{H,v} \quad (\text{S13})$$

$$\tilde{N}_t^{H,v} + \tilde{N}_t^{H,d} = c_3 \dot{N}_t^{H,v} + c_3 \dot{N}_t^{H,d} \quad (\text{S14})$$

Equations S9, S11, S13, and S14 are linear in $v_t := (d_t, N_t, \dot{N}_t, \tilde{N}_t)$ and can be written in matrix form as

$$\underbrace{\begin{pmatrix} 0 & -c_2 & -c_2 & 0 & 0 & 0 & 0 & 0 & 0 & c_2/c_1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & -1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -c_3 & -c_3 & 0 & 0 & 0 & 0 & 1 & 1 \end{pmatrix}}_A \begin{pmatrix} d_t \\ N_t^{P,v} \\ N_t^{P,d} \\ N_t^{I,v} \\ N_t^{I,d} \\ N_t^{H,v} \\ N_t^{H,d} \\ \dot{N}_t^{H,v} \\ \dot{N}_t^{H,d} \\ \tilde{N}_t^{P,v} \\ \tilde{N}_t^{P,d} \\ \tilde{N}_t^{I,v} \\ \tilde{N}_t^{I,d} \\ \tilde{N}_t^{H,v} \\ \tilde{N}_t^{H,d} \end{pmatrix} = 0,$$

or more compactly, $Av_t = 0$ for $t = 1, \dots, T$.

Equations S8, S10, and S12 can be rearranged to group the convex and concave terms separately:

$$\begin{aligned} \underbrace{(\alpha_P d_t + \beta_P d_t^2 - \log N_t^{P,v} - \log c_1)}_{g^P(v_t)} + \underbrace{\log \tilde{N}_t^{P,v}}_{h^P(v_t)} &= 0 \\ \underbrace{(\alpha_I d_t + \beta_I d_t^2 - \log N_t^{I,v})}_{g^I(v_t)} + \underbrace{\log \tilde{N}_t^{I,v}}_{h^I(v_t)} &= 0 \\ \underbrace{(\alpha_H d_t + \beta_H d_t^2 - \log N_t^{H,v})}_{g^H(v_t)} + \underbrace{\log \dot{N}_t^{H,v}}_{h^H(v_t)} &= 0, \end{aligned}$$

where g^X is convex and h^X is concave for every $X \in \{P, I, H\}$. We replace each equality constraint

with two inequality constraints $\leq$ and $\geq$, transforming these equations into

$$\underbrace{\begin{pmatrix} g^P(v_t) \\ g^I(v_t) \\ g^H(v_t) \\ -h^P(v_t) \\ -h^I(v_t) \\ -h^H(v_t) \end{pmatrix}}_{g(v_t)} + \underbrace{\begin{pmatrix} h^P(v_t) \\ h^I(v_t) \\ h^H(v_t) \\ -g^P(v_t) \\ -g^I(v_t) \\ -g^H(v_t) \end{pmatrix}}_{h(v_t)} \leq 0$$

with g convex and h concave. Thus, we have shown that the pre-recompartmentalization cell dynamics S25–S30 can be written equivalently as $Av_t = 0$ and $g(v_t) + h(v_t) \leq 0$ for $t = 1, \dots, T$.

A.2 Post-recompartmentalization

Let $\tilde{N}_t^{tot} = \tilde{N}_t^P + \tilde{N}_t^I + \tilde{N}_t^H$ be the total number of cells just prior to recompartmentalization, but after radiation cell-kill, proliferation, and cell loss take place in time step t . Once recompartmentalization is completed, the cells must be in one of three mutually exclusive configurations. We represent each configuration as a set of constraints on the post-recompartmentalization cell counts $N_{t+1} = (N_{t+1}^{P,v}, N_{t+1}^{P,d}, N_{t+1}^{I,v}, N_{t+1}^{I,d}, N_{t+1}^{H,v}, N_{t+1}^{H,d})$ for $t = 1, \dots, T$.

1. All cells lie in the P compartment, while the I and H compartments are empty.

$$\begin{aligned} \mathcal{C}_{t,1} = \{ (N_{t+1}, \tilde{N}_t) : \tilde{N}_t^{tot} \leq N_1^P, N_{t+1}^{P,v} + N_{t+1}^{P,d} = \tilde{N}_t^{tot}, \\ N_{t+1}^{I,v} + N_{t+1}^{I,d} = 0, N_{t+1}^{H,v} + N_{t+1}^{H,d} = 0 \} \end{aligned} \quad (\text{S15})$$

2. The P compartment is full, and any excess cells lie only in the I compartment, while the H compartment remains empty.

$$\begin{aligned} \mathcal{C}_{t,2} = \{ (N_{t+1}, \tilde{N}_t) : N_1^P \leq \tilde{N}_t^{tot} \leq N_1^P + N_1^I, N_{t+1}^{P,v} + N_{t+1}^{P,d} = N_1^P, \\ N_{t+1}^{I,v} + N_{t+1}^{I,d} = \tilde{N}_t^{tot} - N_1^P, N_{t+1}^{H,v} + N_{t+1}^{H,d} = 0 \} \end{aligned} \quad (\text{S16})$$

3. The P and I compartments are full, and any excess cells spill over into the H compartment.

$$\begin{aligned} \mathcal{C}_{t,3} = \{ (N_{t+1}, \tilde{N}_t) : \tilde{N}_t^{tot} \geq N_1^P + N_1^I, N_{t+1}^{P,v} + N_{t+1}^{P,d} = N_1^P, \\ N_{t+1}^{I,v} + N_{t+1}^{I,d} = N_1^I, N_{t+1}^{H,v} + N_{t+1}^{H,d} = \tilde{N}_t^{tot} - N_1^P - N_1^I \} \end{aligned} \quad (\text{S17})$$

Thus, if at the end of time step t / beginning of time step $t + 1$, the dose delivered up to that point $(d_1, \dots, d_t)$ results in all cells belonging to the P compartment, we must have $(N_{t+1}, \tilde{N}_t) \in \mathcal{C}_{t,1}$.

The cell counts $(N_{t+1}, \tilde{N}_t, \dot{N}_t)$ are state variables that depend on the control variable (i.e., dose schedule) $d = (d_1, \dots, d_T)$. Since we do not know a priori the optimal d , we do not know the corresponding cell counts or optimal cell configuration at time t , and so cannot say a priori whether to constrain $(N_{t+1}, \tilde{N}_t) \in \mathcal{C}_{t,s}$ for $s = 1, 2$, or 3 . This determination must be made during optimization, simultaneously with the dose schedule. In other words, we must define the optimal dose scheduling problem such that its feasible set allows for any one of the three cell

configurations to be active at time $t = 1, \dots, T$. We do this by introducing a new state variable $z_t \in \{0, 1\}^3$. The binary $z_{t,s}$ indicates whether cells are in configuration s after recompartimentalization in time step t , i.e., $z_{t,s} = 1$ if and only if $(\tilde{N}_t, N_{t+1}) \in \mathcal{C}_{t,s}$. Since the cell configurations are mutually exclusive, we must have $z_{t,1} + z_{t,2} + z_{t,3} = 1$ (this restricts $z_{t,s} = 1$ for exactly one s).

Now, let $R \gg 0$ be a constant, and consider the constraint set

$$\begin{aligned} \mathcal{S}_{t,1}(z_{t,1}) = \{ & (N_{t+1}, \tilde{N}_t) : \tilde{N}_t^{tot} - N_1^P - (1 - z_{t,1})R \leq 0, \\ & N_{t+1}^{P,v} + N_{t+1}^{P,d} - \tilde{N}_t^{tot} - (1 - z_{t,1})R \leq 0, \\ & -N_{t+1}^{P,v} - N_{t+1}^{P,d} + \tilde{N}_t^{tot} - (1 - z_{t,1})R \leq 0 \\ & N_{t+1}^{I,v} + N_{t+1}^{I,d} - (1 - z_{t,1})R \leq 0, \\ & -N_{t+1}^{I,v} - N_{t+1}^{I,d} - (1 - z_{t,1})R \leq 0 \\ & N_{t+1}^{H,v} + N_{t+1}^{H,d} - (1 - z_{t,1})R \leq 0, \\ & -N_{t+1}^{H,v} - N_{t+1}^{H,d} - (1 - z_{t,1})R \leq 0 \} \end{aligned} \quad (\text{S18})$$

If $z_{t,1} = 1$, all the $(1 - z_{t,1})R$ terms disappear, and the set is identical to S15 ($\mathcal{S}_{t,1}(1) = \mathcal{C}_{t,1}$). If $z_{t,1} = 0$, we can choose R large enough such that all the inequalities in S18 hold automatically for every realistic value of the cell counts $(N_{t+1}, \tilde{N}_t)$ over $t = 1, \dots, T$. (We will discuss how to choose such an R later in this subsection). Then, the constraint $(N_{t+1}, \tilde{N}_t) \in \mathcal{S}_{t,1}(0)$ can be effectively ignored. Putting everything together, we conclude that the logical statement “ $z_{t,1} = 1$ if and only if $(N_{t+1}, \tilde{N}_t) \in \mathcal{C}_{t,1}$ ” is equivalent to “ $(N_{t+1}, \tilde{N}_t) \in \mathcal{S}_{t,1}(z_{t,1})$ for $z_{t,1} \in \{0, 1\}$.”

The above argument holds for S16 and S17 as well, with the corresponding constraint sets

$$\begin{aligned} \mathcal{S}_{t,2}(z_{t,2}) = \{ & (N_{t+1}, \tilde{N}_t) : -\tilde{N}_t^{tot} + N_1^P - (1 - z_{t,2})R \leq 0, \\ & \tilde{N}_t^{tot} - N_1^P - N_1^I - (1 - z_{t,2})R \leq 0, \\ & N_{t+1}^P + N_{t+1}^{P,d} - N_1^P - (1 - z_{t,2})R \leq 0, \\ & -N_{t+1}^{P,v} - N_{t+1}^{P,d} + N_1^P - (1 - z_{t,2})R \leq 0, \\ & N_{t+1}^{I,v} + N_{t+1}^{I,d} - \tilde{N}_t^{tot} + N_1^P - (1 - z_{t,2})R \leq 0, \\ & -N_{t+1}^{I,v} - N_{t+1}^{I,d} + \tilde{N}_t^{tot} - N_1^P - (1 - z_{t,2})R \leq 0, \\ & N_{t+1}^{H,v} + N_{t+1}^{H,d} - (1 - z_{t,2})R \leq 0, \\ & -N_{t+1}^{H,v} - N_{t+1}^{H,d} - (1 - z_{t,2})R \leq 0 \} \end{aligned} \quad (\text{S19})$$

$$\begin{aligned} \mathcal{S}_{t,3}(z_{t,3}) = \{ & (N_{t+1}, \tilde{N}_t) : -\tilde{N}_t^{tot} + N_1^P + N_1^I - (1 - z_{t,3})R \leq 0, \\ & N_{t+1}^{P,v} + N_{t+1}^{P,d} - N_1^P - (1 - z_{t,3})R \leq 0, \\ & -N_{t+1}^{P,v} - N_{t+1}^{P,d} + N_1^P - (1 - z_{t,3})R \leq 0, \\ & N_{t+1}^{I,v} + N_{t+1}^{I,d} - N_1^I - (1 - z_{t,3})R \leq 0, \\ & -N_{t+1}^{I,v} - N_{t+1}^{I,d} + N_1^I - (1 - z_{t,3})R \leq 0, \\ & N_{t+1}^{H,v} + N_{t+1}^{H,d} - \tilde{N}_t^{tot} + N_1^P + N_1^I - (1 - z_{t,3})R \leq 0, \\ & -N_{t+1}^{H,v} - N_{t+1}^{H,d} + \tilde{N}_t^{tot} - N_1^P - N_1^I - (1 - z_{t,3})R \leq 0 \} \end{aligned} \quad (\text{S20})$$

Thus, the full post-recompartmentalization cell dynamics can be represented compactly as

$$\begin{aligned} (N_{t+1}, \tilde{N}_t) &\in \mathcal{S}_{t,s}(z_{t,s}), \quad s = 1, 2, 3, \\ z_t &\in \{0, 1\}^3, \quad z_{t,1} + z_{t,2} + z_{t,3} = 1 \end{aligned}$$

for $t = 1, \dots, T$.

It remains to select the constant $R \gg 0$. We want a value of R such that if $z_{t,1} = 1$, all the inequalities in S18 automatically hold for any realistic value of $(N_{t+1}, \tilde{N}_t)$, and we can ignore the constraint $(N_{t+1}, \tilde{N}_t) \in \mathcal{S}_{t,1}(1)$. First, consider $\tilde{N}_t^{tot} - N_1^P - R \leq 0$. Since $N_1^P \geq 0$, it suffices to find an upper bound $R \geq N_t^{tot}$, i.e., R must be larger than the total number of cells in time step t after proliferation, cell loss/death, and radiation cell-kill take place. All of these processes except proliferation result in a reduction of the number of viable cells. Thus, a possible R is the total number of cells at the end of the final time step T if cellular resources are unlimited and all cells undergo only proliferation (this is biologically impossible, but produces a reliable upper bound, which is all we need):

$$R = N_1^{tot} \exp \left(T f_{pro}^P \frac{\log 2}{T_C} \Delta t \right). \quad (\text{S21})$$

By construction, we have $R \geq \tilde{N}_t^{tot} \geq \tilde{N}_t^{tot} - N_1^P$ for all $t = 1, \dots, T$, so the first inequality in S18 is satisfied when $z_{t,1} = 0$. In fact, all the inequalities are satisfied because in any time step t , R is larger than the total number of cells, so it is larger than the total number of cells in a compartment, i.e., $R \geq \sum_{X \in \{P, I, H\}} N_{t+1}^{X,v} + N_{t+1}^{X,d} \geq N_{t+1}^{X,v} + N_{t+1}^{X,d}$ for every $X \in \{P, I, H\}$. Hence, when $z_{t,1} = 0$, the constraint $(N_{t+1}, \tilde{N}_t) \in \mathcal{S}_{t,1}(z_{t,1})$ holds for all realistic values of $(N_{t+1}, \tilde{N}_t)$ and is effectively ignored during optimization. The same argument can be applied to $(N_{t+1}, \tilde{N}_t) \in \mathcal{S}_{t,2}(z_{t,2})$ and $(N_{t+1}, \tilde{N}_t) \in \mathcal{S}_{t,3}(z_{t,3})$ with the same definition of R given in S21.

Since every set $\mathcal{S}_{t,s}(z_{t,s})$ contains only mixed-integer (binary) linear inequality constraints, we can rewrite $(\tilde{N}_t, N_{t+1}) \in \mathcal{S}_{t,1}(z_{t,s})$, $s = 1, 2, 3$ as a single mixed-integer linear matrix inequality

constraint

$$\underbrace{\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 \\ -1 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & -1 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & -1 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 \\ -1 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & -1 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & -1 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 \\ -1 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & -1 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & -1 & -1 \end{pmatrix}}_F \begin{pmatrix} N_{t+1}^{P,v} \\ N_{t+1}^{P,d} \\ N_{t+1}^{I,v} \\ N_{t+1}^{I,d} \\ N_{t+1}^{H,v} \\ N_{t+1}^{H,d} \end{pmatrix} + \underbrace{\begin{pmatrix} 1 \\ -1 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ -1 \\ 1 \\ 0 \\ -1 \\ 1 \\ 0 \\ 0 \\ -1 \\ 0 \\ 0 \\ 0 \\ -1 \\ 1 \end{pmatrix}}_{\tilde{H}} \tilde{N}_t^{tot} - \underbrace{\begin{pmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{pmatrix}}_G z_t + \underbrace{\begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ N_1^P \\ -N_1^P - N_1^I \\ -N_1^P \\ N_1^P \\ N_1^P \\ -N_1^P \\ 0 \\ 0 \\ N_1^P + N_1^I \\ -N_1^P \\ N_1^P \\ -N_1^I \\ N_1^I \\ N_1^I \\ N_1^P + N_1^I \\ -N_1^P - N_1^I \end{pmatrix}}_{\tilde{c}} \leq R,$$

or simply $FN_{t+1} + \tilde{F}\tilde{N}_t + Gz_t + c \leq 0$ with $\tilde{F} = \tilde{H}\mathbf{1}^T$ (since $\tilde{N}_t^{tot} = \sum_{X \in \{P,I,H\}} \tilde{N}_t^{X,v} + \tilde{N}_t^{X,d}$) and $c = \tilde{c} - R\mathbf{1}$.

We are now ready to define the optimal dose scheduling problem 9. Let (N_1^P, N_1^I, N_1^H) be the initial number of cells in each compartment, and T be the total length of treatment represented as the number of time steps Δt . The control variable is the dose schedule $d = (d_1, \dots, d_T)$, and the state variables are the cell counts, $N = (N_2, \dots, N_{T+1})$, $\dot{N} = (\dot{N}_1, \dots, \dot{N}_T)$, and $\tilde{N} = (\tilde{N}_1, \dots, \tilde{N}_T)$, and the cell configuration indicator $z = (z_1, \dots, z_T)$. All variables must be nonnegative, and z_t in particular must be binary with $z_{t,1} + z_{t,2} + z_{t,3} = 1$ for $t = 1, \dots, T$.

Our goal is to minimize the fraction of surviving tumor cells at the end of treatment, $f_0(N_{t+1}) = \mathbf{1}^T N_{T+1} / \mathbf{1}^T N_1$, subject to an upper bound on the normal tissue BED₃, $f_1(d) = \sum_{t=1}^T (d_t + \beta_N / \alpha_N d_t^2) \leq B^{\max}$. In time step t , the pre-recompartmentalization cell dynamics are represented by the constraints $Av_t = 0$ and $g(v_t) + h(v_t) \leq 0$, where $v_t := (d_t, N_t, \dot{N}_t, \tilde{N}_t)$, and the post-recompartmentalization cell counts are determined by the constraint $f_2(N_{t+1}, \tilde{N}_t, z_t) = FN_{t+1} + \tilde{F}\tilde{N}_t + Gz_t + c \leq 0$. Putting it all together, our optimal control problem is

$$\begin{aligned}
 & \text{minimize} && f_0(N_{T+1}) \\
 & \text{subject to} && f_1(d) \leq B^{\max} \\
 & && Av_t = 0, \quad g(v_t) + h(v_t) \leq 0, \quad t = 1, \dots, T \\
 & && f_2(N_{t+1}, \tilde{N}_t, z_t) \leq 0, \quad t = 1, \dots, T \\
 & && d \geq 0, \quad N \geq 0, \quad \dot{N} \geq 0, \quad \tilde{N} \geq 0 \\
 & && z_t \in \{0, 1\}^3, \quad z_{t,1} + z_{t,2} + z_{t,3} = 1
 \end{aligned}$$

with respect to $d, N, \dot{N}, \tilde{N}$, and z .

B Tumor Dose-Response Model

We briefly review the tumor dose-response model presented in [1], which provides the dynamics for our optimal dose scheduling problem. This model captures the interaction between several important radiobiological phenomena, including cellular proliferation, radiation cell-kill, competition for limited oxygen and glucose, and hypoxia-induced radioresistance. Cells reside in one of three conceptual compartments: proliferative cells with access to both oxygen and glucose (P compartment), intermediately resource-constrained cells with access to glucose but not oxygen (I compartment), and extremely hypoxic cells without access to oxygen or glucose (H compartment). The initial number of cells in each compartment is determined by the growth fraction GF and the cell loss factor CLF , i.e., the rate of cell loss divided by the rate of proliferation in the tumor:

$$N_1^P = \frac{GF}{f_{pro}^P} \cdot N_1^{tot} \quad (S22)$$

$$N_1^I = \left(1 - GF \left(\frac{1}{f_{pro}^P} + CLF \cdot \frac{T_{1/2,loss}}{T_C} \right) \right) N_1^{tot} \quad (S23)$$

$$N_1^H = CLF \cdot GF \cdot \frac{T_{1/2,loss}}{T_C} \times N_1^{tot}, \quad (S24)$$

where N_1^{tot} is the total number of cells in the tumor, f_{pro}^P is the fraction of cells in the P compartment that are in cell cycle at any given time, T_C is the cell cycle time, and $T_{1/2,loss}$ is the cell loss half-time [1, Equation A.4].

Let N_t^X be the number of cells in compartment X at the beginning of the t -th discrete time step of length Δt , where $t \in \{1, 2, \dots\}$ and typically $\Delta t \approx 15$ minutes or less. Within each time step, several radiobiological processes take place that affect the distribution of cells across compartments.

Radiation cell-kill. During radiotherapy (RT), radiation dooms a fraction of cells in each compartment. These doomed cells are lethally damaged, so do not proliferate, but survive metabolically until they die during failed mitosis. We follow the linear-quadratic (LQ) model [2] in calculating radiation damage: for a given compartment $X \in \{P, I, H\}$, the number of viable (non-doomed) cells and the number of doomed cells left after a fractional dose $d_t \geq 0$ are, respectively,

$$N_{t+1}^{X,v} = N_t^{X,v} \exp(-\alpha_X d_t - \beta_X d_t^2), \quad (S25)$$

$$N_{t+1}^{X,d} = N_t^{X,d} + N_t^{X,v} (1 - \exp(-\alpha_X d_t - \beta_X d_t^2)), \quad (S26)$$

where $\alpha_X > 0$ and $\beta_X > 0$ are the LQ model parameters for cells in compartment X . Hypoxic cells are typically less radiosensitive than normoxic cells, which we express through an oxygen enhancement ratio (OER) as follows [3]:

$$\alpha_X = \alpha_P / OER_X, \quad \beta_X = \beta_P / OER_X^2. \quad (S27)$$

Proliferation and cell loss. The P compartment receives enough oxygen and glucose to support a fraction f_{pro}^P of cells in cell cycle. Thus, proliferation results in the number of viable cells becoming

$$N_{t+1}^{P,v} = N_t^{P,v} \exp \left(f_{pro}^P \frac{\log 2}{T_C} \Delta t \right). \quad (\text{S28})$$

On the other hand, the H compartment has insufficient oxygen and glucose, which leads to overall cell loss according to [4]

$$N_{t+1}^H = N_t^H \exp \left(-\frac{\log 2}{T_{1/2,loss}} \Delta t \right), \quad (\text{S29})$$

where $T_{1/2,loss}$ is the cell loss half-time. The I compartment receives adequate glucose but not oxygen, so its cells neither proliferate nor undergo cell loss due to starvation, remaining in a metabolically active state outside the cell cycle.

Recompartmentalization and reoxygenation. Doomed cells in the P compartment die probabilistically following mitosis. Mitotic death may not occur on the first attempt post-damage division, but require several attempts, with the survival probability of each progeny typically being $k_m \in [0.1, 0.4]$ [5]. Therefore, the number of doomed cells evolve according to

$$N_{t+1}^{P,d} = N_t^{P,d} \exp \left((2k_m - 1) f_{pro}^P \frac{\log 2}{T_C} \Delta t \right). \quad (\text{S30})$$

Cell death frees up chemical resources, i.e., oxygen and glucose. Cells in the I compartment gain access to the oxygen and move into the P compartment, in turn liberating resources so cells in the H compartment can move into the I compartment. This process of recompartmentalization and reoxygenation of hypoxic cells is illustrated in Figure S1.

C Convergence of Algorithm 1 without slack

The following proof is based on [6, Section 1.3]. We are interested in analyzing the convergence of Algorithm 1 when no slack is allowed in the approximation of $h(v_t)$, i.e., the slack term $\delta := 0$ and its associated penalty $r(\delta) := 0$. Under these conditions, we first show that all iterates of the algorithm are feasible. Let v^k be a feasible point of the original nonconvex problem 9, then v^k is also feasible for the MICP approximation problem 12 because

$$g(v_t^k) + \hat{h}(v_t^k; v_t^k) = g(v_t^k) + h(v_t^k) \leq 0, \quad t = 1, \dots, T,$$

where we have simply applied definition 11. Thus, problem 12 has a feasible point v^{k+1} . Moreover, since h is concave, we have $h(u) \leq \hat{h}(u; v_t^k)$ for all u , and

$$g(v_t^{k+1}) + h(v_t^{k+1}) \leq g(v_t^{k+1}) + \hat{h}(v_t^{k+1}; v_t^k) \leq 0, \quad t = 1, \dots, T.$$

Thus, v^{k+1} is feasible for the original problem 9. Since v^0 is assumed to be feasible, this implies that all iterates are feasible.

Now, consider the objective value $f_0(N_{T+1}^{k+1})$. From Algorithm 1, $v^{k+1} := (d_t^{k+1}, N_t^{k+1}, \dot{N}_t^{k+1}, \tilde{N}_t^{k+1})$ is an optimal point of MICP problem 12, therefore we must have $f_0(N_{T+1}^{k+1}) \leq f_0(N_{T+1})$ for all

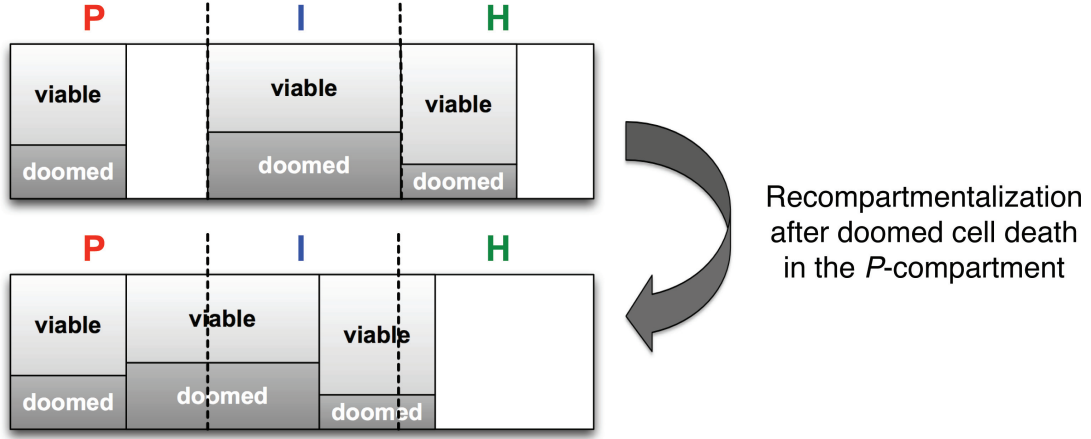


Figure S1: After dead cells are removed from the tumor, reoxygenation occurs as hypoxic cells in the I and H compartments gain access to oxygen and glucose. This recompartmentalization process can be visualized as cells in the I compartment filling up the freed space (representing newly available resources) in the P compartment and cells in the H compartment filling up the freed space in the I compartment.

feasible $v := (d_t, N_t, \dot{N}_t, \tilde{N}_t)$ of problem 12. In particular, since $v^k := (d_t^k, N_t^k, \dot{N}_t^k, \tilde{N}_t^k)$ is a feasible point of problem 12, it must be that

$$p_{opt}^{k+1} = f_0(N_{T+1}^{k+1}) \leq f_0(N_{T+1}^k) = p_{opt}^k, \quad k = 1, 2, \dots$$

Thus, the sequence of optimal objective values $\{p_{opt}^k\}_{k=1}^{\infty}$ is nonincreasing and must converge, possibly to $-\infty$. (Note: While convergence to a local minimum is not guaranteed, in certain cases, convergence to a critical point can be proven [7]).

D Computational Implementation

All computational experiments were carried out on a 64-bit Windows 10 desktop with a 12-core AMD Ryzen 9 3900X / 3.80 GHz CPU and 128 GB of RAM. The parameters used in our experiments are given in Supplementary Table S2. Initial cell counts (N_1^P, N_1^I, N_1^H) and LQ model parameters $(\alpha_I, \beta_I, \alpha_H, \beta_H)$ were calculated from equations S22, S23, and S24 and equation S27, respectively. In addition to the constraints described in Section 2.1, we also enforced an upper bound on the dose delivered per day of $D^{\max} = 18$ Gy, which we incorporated into problem 9 via a simple linear constraint on d . This choice of $D^{\max}$ along with the choices of $B^{\max}$ in our simulations were derived from common NSCLC fractionation schedules described in Gouw et al.[8] (e.g., $7.5 \text{ Gy} \times 8$ yields a BED_3 of 210 Gy, which we used as $B^{\max}$ in generating Figure 1). To enforce a constant dose constraint, we introduced a new variable $d^{\text{const}} \geq 0$ and added the constraint $d_t = d^{\text{const}}$ for all $t = 1, \dots, T$ to our optimization problem. If we wished to enforce a treatment break, we set $d_t = 0$ for all $t \in \mathcal{T}_B$, where $\mathcal{T}_B$ is the set of time steps corresponding to the length and location of the break.

D.1 Initialization

As a local heuristic, the output of Algorithm 1 may often depends on the initial point v^0 . It is thus good practice to run the algorithm with several different (feasible) v^0 and take as the final result the output corresponding to the lowest objective value. In our computations, we generated the initial dose d^0 by uniformly sampling from the nonnegative reals up to a maximum dose, then calculated the cell counts in each compartment using the tumor response model (Supplementary Section B) to obtain v^0 .

More precisely, we computed d^0 as follows: given an upper bound $B^{\max}$ on the normal tissue BED₃, we determined the corresponding maximum allowable dose assuming the dose fraction d_t is constant, i.e., we solved the quadratic equation $f_1((d^c, \dots, d^c)) - B^{\max} = 0$ for $d^c \in \mathbf{R}_+$ and set $d^0 = (d^c, \dots, d^c)$. Then, we plugged d^0 into the reformulated cell dynamics equations in Supplementary Section S1–S7 to obtain the resulting cell counts $N^0, \dot{N}$, and $\tilde{N}$. The initial point $v^0 := (d^0, N^0, \dot{N}^0, \tilde{N}^0)$.

D.2 Speed considerations

For large problems, the runtime of Algorithm 1 may become prohibitive due to the size of the mixed-integer constraint 8. We suggest first solving problem 9 without this constraint, which can be accomplished via standard (non-mixed-integer) CCP, then using its result as a warm start for the full problem. We found in our experiments that the first step already produces a fairly good dose schedule, such that the second step acts as a minor refinement, requiring only 1–3 iterations of Algorithm 1.

E Additional results and discussion

E.1 Enforcement of a treatment break constraint

We designed Algorithm 1 to find the dose schedule that achieves the highest TCP, but in the clinic, it is equally important that such a schedule be practically deliverable. In our initial computational experiments, we discovered that without a constraint on the search space, the algorithm tended to produce dose schedules with many consecutive small fractions (typically ≤ 1 Gy), which contribute minimally to reducing TCP, yet are impractical to deliver in a clinical setting. Thus, we made the decision to restrict the search space by enforcing a treatment break right after the primer shot as proposed by Gouw et al. [8]. We also verified that, all else equal, the optimal dose schedule with the treatment break constraint achieves nearly the same TCP as the optimal schedule without this constraint. For instance, in our experiments that produced Figure 1c, the dose schedule with an enforced break yields a TCP of 0.9321, while the schedule without a break yields a TCP of 0.9375, a difference on the order of only 10^{-3} . Results are similar for the other figures. Broadly speaking, Algorithm 1 should be considered a *tool* for exploring dose schedules under a variety of different constraints, with the clinician making the final decision on which constraints to enforce and subsequent schedule to deliver in a particular patient case.

E.2 Comparison with best constant fractionation schedules

We assess how close our sequential MICP approximations come to capturing the impact of the nonconvex cell dynamics by comparing five optimal dose schedules generated by Algorithm 1

(Table S3) with dose schedules found via a brute force search. To make this search tractable, we restricted our search space to schedules of constant fraction size with the same maximum treatment length $T^{\max}$ and normal tissue BED_3 as the algorithmically generated schedules, meaning we only needed to iterate over all possible combinations of treatment days/breaks. For each set of parameter values ($T^{\max}$ and normal tissue BED_3), our search identified the dose schedule with the lowest TCP as computed by the tumor dose-response model described in Supplementary Section B. The results are shown in Table S4.

Using this approach, we are able to draw a fair comparison between the schedule generated by Algorithm 1 and the schedule returned by a brute force search with the same parameter values. In general, we see that the TCP of every schedule produced by our algorithm is within less than 0.01 of the TCP of the analogous schedule found via brute force search with the same normal tissue BED_3 . Similarly, the EQD2 values are within only 2 Gy of each other. The main disparity between the dose schedules returned by the two methods is that the brute force search produced schedules with fewer fractions, but larger fraction size. For example, with $T^{\max} = 16$ and normal tissue $BED_3 = 202.95$, Algorithm 1 gave us a schedule with 8 fractions, an average dose of 6.55 Gy, and a maximum dose of 16.27 Gy, spread out evenly after the reoxygenation break. By contrast, the best constant fractionation schedule found via brute force search delivers 4 fractions of 10.93 Gy, with the first being a primer shot and the rest concentrated near the end of treatment (days 12, 15, and 16). Overall, these results suggest that the optimal dose scheduling problem potentially has multiple solutions, with both constant and variable fractionation schedules capable of attaining the minimum tumor SF subject to the dose constraints.

E.3 Optimizing with respect to treatment break length

It is possible to include the length of the treatment break as a variable in the optimal dose scheduling problem 9. Let us assume for simplicity that each time step t corresponds to a potential treatment day, with the maximum number of days being T . Define control variables $\psi_t \in \{0, 1\}$ for $t = 1, \dots, T$, where $\psi_t = 0$ if a treatment break is to be enforced on day t and $\psi_t = 1$ otherwise. We then enforce a maximum dose constraint $d_t \leq \psi_t D^{\max}$ for $t = 1, \dots, T$, so that $d_t = 0$ when $\psi_t = 0$. The total number of break days, $\sum_{t=1}^T \psi_t$, can then be incorporated into the scheduling problem, either as part of the objective or additional constraints, e.g., we may require $\Psi^{\min} \leq \sum_{t=1}^T \psi_t \leq \Psi^{\max}$, where $\Psi^{\min} > 0$ and $\Psi^{\max} > 0$ are user-specified minimum and maximum number of break days, respectively.

From a mathematical perspective, this is a straightforward augmentation of the original problem. The variables $\{\psi_t\}_{t=1}^T$ are binary and the associated constraints are linear, so the augmented problem is MICP and Algorithm 1 may be applied. However, we found in our experiments that the additional control variables add considerable computational time to the optimization, and it is more efficient to simply re-solve the original problem with different length breaks manually enforced over a fixed time period, then compare the resulting schedules. One could start with a short break, which is extended day by day until complete reoxygenation occurs, or use a divide-and-conquer search method. Developing such a method for efficiently optimizing treatment break time is a direction for future research.

F Supplementary Tables

F.1 Optimization problem variables and parameters

Table S1: Optimal dose scheduling problem variables and parameters (Section 2.1)

Symbol	Domain	Description
T	$\mathbf{N}$	Total number of time steps
d_t	$\mathbf{R}_+$	Dose fraction in time step t
$N_t^{X,v}$	$\mathbf{R}_+$	Number of viable cells in compartment $X \in \{P, I, V\}$ at the beginning of time step t
$N_t^{X,d}$	$\mathbf{R}_+$	Number of doomed cells in compartment $X \in \{P, I, V\}$ at the beginning of time step t
$\dot{N}_t^{H,v}$	$\mathbf{R}_+$	Number of viable cells in compartment H right after radiation cell-kill in time step t
$\dot{N}_t^{H,d}$	$\mathbf{R}_+$	Number of doomed cells in compartment H right after radiation cell-kill in time step t
$\tilde{N}_t^{X,v}$	$\mathbf{R}_+$	Number of viable cells in compartment $X \in \{P, I, V\}$ just before recompartmentalization in time step t
$\tilde{N}_t^{X,d}$	$\mathbf{R}_+$	Number of doomed cells in compartment $X \in \{P, I, V\}$ just before recompartmentalization in time step t
$z_{t,s}$	$\{0, 1\}$	Indicator that recompartmentalized cells are in configuration $s \in \{1, 2, 3\}$ at the end of time step t (see constraints 5—7)
δ_t	$\mathbf{R}_+$	Slack penalty on convex approximation of dynamics equation 11 in time step t
N_1^{tot}	$\mathbf{R}_+$	Total number of tumor cells before the start of treatment
α_N	$\mathbf{R}_{++}$	Normal tissue LQ model α parameter
β_N	$\mathbf{R}_{++}$	Normal tissue LQ model β parameter
A	$\mathbf{R}_+^{m \times n_x}$	Dose-influence matrix (m = number of voxels, n_x = number of beamlets)
C	$\mathbf{R}^{K \times T}$	(Optional) Linear inequality constraint matrix on doses d (K = number of constraints)
b	$\mathbf{R}^K$	(Optional) Linear inequality constraint upper bound on doses d (i.e., $Cd \leq b$)
λ	$\mathbf{R}_{++}$	Slack penalty function weight parameter

Table S2: Model parameters¹ used in computational simulations (Section 3)

Parameter	Value	Description
ρ_t	10^6 mm^{-3}	Tumor cell density
v_t	64 mm^3	Volume of a tumorlet (typical PET voxel size)
N_1^{tot}	6.4×10^7	Total number of cells in a tumorlet ($\rho_t v_t$)
CLF	0.92	Tumor cell loss factor
GF	0.25	Tumor growth fraction
T_C	2 days	Cell cycle time
f_{pro}^P	0.5	Initial proliferation fraction in P compartment
$T_{1/2,loss}$	2 days	Cell loss half-time in H compartment
$T_{1/2,lysis}$	3 days	Lysis half-time
k_m	0.3	Survival rate of progeny after mitosis
α_P	0.305	α for tumor cells in P compartment
α_P / β_P	2.8	α / β ratio for tumor cells in P compartment
OER_I	1.7	Oxygen enhancement ratio for I compartment
OER_H	1.37	Oxygen enhancement ratio for H compartment
α_N / β_N	3	α / β ratio for normal tissue
Δt	60 secs ²	Time elapsed in each time step
T	Varies ³	Total number of time steps
λ	$1/T$	Slack penalty function weight
$D^{\max}$	18 Gy	Upper bound on dose per day
$B^{\max}$	Varies	Upper bound on normal tissue BED ₃

F.2 Model parameters for simulations

F.3 Dose schedules generated by Algorithm 1 vs. brute force search

¹The tumor, cell compartment, and radiosensitivity parameters are adapted from Table 1 in Jeong et al. [1]

²The integration time step Δt is used in the dynamical system to resolve the immediate post-irradiation evolution of the state variables. Since the reoxygenation component is nonlinear due to its coupling with the viable cell count, we selected a relatively small $\Delta t = 60$ secs to promote accuracy and numerical stability of Algorithm 1.

³If T^d is the total treatment length in days, we calculate $T = (T^d \times 24 \times 60 \times 60) / \Delta t$ with $\Delta t = 60$ secs. In our simulations, T^d varied between 10 and 19 days. See Section 3 of the main paper for details.

Table S3: Optimal dose schedules with variable fraction size using Algorithm 1

Schedule	# Fractions	Avg. Dose	Max. Dose	TCP	EQD2	Normal BED ₃
1–5, 8–10	8	6.26	17.47	0.8743	93.37	207.04
1, 8–12, 15–16	8	5.50	14.60	0.8827	95.37	156.77
1, 8–12, 15–16	8	9.88	17.97	0.9493	207.87	371.14
1, 8–12, 15–16	8	6.55	16.27	0.9321	119.98	202.95
1, 8–12, 15–19	11	5.27	15.80	0.9412	135.42	208.14

Table S4: Best dose schedules with constant fraction size using brute force search

Schedule	# Fractions	$T^{\max}$	Dose	TCP	EQD2	Normal BED ₃
1–5, 8–10	8	10	7.44	0.8826	95.33	207.04
1, 3, 11, 12, 15, 16	6	16	7.48	0.8751	93.54	156.77
1, 12, 15, 16	4	16	15.25	0.9494	209.59	371.14
1, 12, 15, 16	4	16	10.93	0.9299	117.68	202.95
1, 3, 11, 12, 15–19	9	19	6.96	0.9421	137.81	208.14

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